

Technical Documentation: Unified Military Analytics data scraping and cleaning(Milestone – 1)

Project: Unified Military Analytics and Visualization (Infosys Internship)

Task: Web Scraping and Data Pipeline Construction

1. Executive Summary

This module facilitates the automated extraction of global military statistics from various web sources. The script implements a robust pipeline that reads source configurations, performs HTTP requests, parses complex HTML structures, cleans numerical data, and aggregates the information into a unified, analysis-ready format.

2. System Architecture

The data pipeline follows a sequential four-stage process:

1. **Configuration Parsing:** Reading target URLs from a local text-based links file(military_data_links.txt).
 2. **Data Acquisition:** Programmatic retrieval of web content using the requests library of python.
 3. **Content Extraction:** Parsing HTML DOM trees via BeautifulSoup library to isolate specific military metrics.
 4. **Data Transformation:** Normalizing numerical strings and structuring the data into a tabular pandas DataFrame, simultaneously treating any null values, missing data or inappropriate data.
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3. Implementation Details

3.1 Dependencies

The script utilizes industry-standard libraries for data science:

- requests: For handling network communication.
- BeautifulSoup (bs4): For HTML parsing and navigation.
- pandas: For data manipulation and structured storage.
- re: For regex-based data cleaning.

3.2 Component Breakdown

A. Dynamic Source Loading (parse_sources)

Instead of hardcoding URLs, the script reads from a configuration file (links_for_military_data.txt). This ensures scalability; new data sources can be added to the text file without modifying the core logic. It specifically filters for sections labeled under other_sources.

B. Targeted Scraping Logic (get_links)

The engine iterates through the identified sources and targets specific CSS classes:

- **Container:** recordsetContainer
- **Country Names:** Extracted from shortFormName and longFormName.
- **Metric Values:** Extracted from valueContainer.

C. Data Cleaning & Normalization

Raw web data is often "noisy." The script applies several cleaning steps:

- **Regex Filtering:** Removes non-numeric characters (commas, currency symbols, or suffixes) using `re.sub(r"[^0-9]", "", cnt_val)`.
- **Type Casting:** Converts cleaned strings into integers for mathematical compatibility.
- **Missing Value Handling:** `dataframe.fillna(0)` to ensure a complete matrix where data might be missing for certain countries.

4. Data Structure

The final output is a Master DataFrame where:

- **Rows:** Represent individual countries (Long-form names).
- **Columns:** Represent different military metrics derived from various sources.
- **Attributes:** Includes a "shortname" for easier mapping in visualization tools.

country shortname Metric_A Metric_B ...

<i>United States</i>	USA	750000	12000	...
<i>India</i>	IND	1450000	4500	...

5. Key Highlights

- **Scalability:** The use of a dictionary-based master_data structure allows for an arbitrary number of metrics to be joined on the "Country" key automatically.
- **Resilience:** The script includes logic to handle inconsistent naming conventions by storing both short and long-form country names.

6. Final Output & Deliverables

After the processing and cleaning pipeline is executed, the system consolidates the disparate data points into a finalized, structured dataset.

- **File Name:** unified_military_data.csv
- **Storage Location:** /content/drive/MyDrive/.../data/processed/
- **Format:** Comma-Separated Values (CSV)

Key Features of the Exported File:

1. **Data Integrity:** All numeric fields are stripped of non-numeric characters (like symbols and commas, units) and converted to standardized integer formats.
2. **Zero-Handling:** Missing values (NaN) from different sources are automatically replaced with 0, ensuring that the dataset is "visualization-ready" and won't cause errors during visualization.
3. **Indexing:** The "Country" attribute is set as the primary key, making it easy to merge with other global datasets or Geographic (GIS) shapefiles.